A case study: the ESICUP/RENAULT competition



Transportation between factories





Very large instances

- Number of required trucks or containers

Instances A	Containers
ARG_BUR_STK_container/	6
AVF_RIR_STK_container/	24
AVF_RIR_STK_truck/	5
AVT_BUR_STK_container/	4
AVZ_RIR_STK_container/	17
AVZ_RIR_STK_truck/	5
CHE_GC_PEX_container/	3
CHE_RIR_STK_container/	12
CME_GC_PEX_container/	5
CVP_GC_PEX_container/	27
CVU_GC_PEX_container/	**85**
FSI_GC_PEX_container/	27
IND_RIR_STK_container/	5
MED_GC_PEX_container/	13
NSA_BUR_STK_container/	5
NSA_RIR_STK_container/	4
RSM_GC_PEX_container/	5
SOM_BUR_STK_container/	**2**
SOM_RIR_STK_container/	26
TNG_RIR_STK_container/	4

Instances B	Containers
ACI_RIR_PEX_truck	4
ALG_RIR_PEX_container	28
AVF_RIR_PEX_container	6
AVZ_RIR_PEX_container	11
CHE_GC_PEX_container	5
CHE_RIR_PEX_container	11
CHE_VLD_PEX_container	9
CVP_GC_PEX_container	7
CVU_GC_PEX_container	22
CVU_VLD_PEX_container	7
FSI_GC_PEX_container	10
FSI_VLD_PEX_container	16
IKO_RIR_PEX_truck	4
MED_GC_PEX_container	8
MJV_VLD_PEX_truck	4
NSA_RIR_PEX_container	18
RSM_GC_PEX_container	7
SOM_VLD_PEX_container	**42**
TAK_RIR_PEX_truck	**2**

Instances X	Containers
ALG_RIR_PEX_container	11
ARG_BUR_PEX_container	14
AVF_RIR_PEX_container	24
AVT_BUR_PEX_container	15
BAX_COR_PEX_container	20
CHE_RIR_PEX_container	15
CME_GC_EXP_container	8
CVP_GC_EXP_container	27
FSI_GC_EXP_container	35
MED_GC_EXP_container	25
NSA_RIR_PEX_container	7
RSM_GC_EXP_container	7
SOM_GC_EXP_container	19
SOM_RIR_PEX_container	32
TAN_BUR_PEX_container	11
TNG_RIR_PEX_container	**54**
VAX_COR_PEX_container	**5**

Two types of containers / trucks

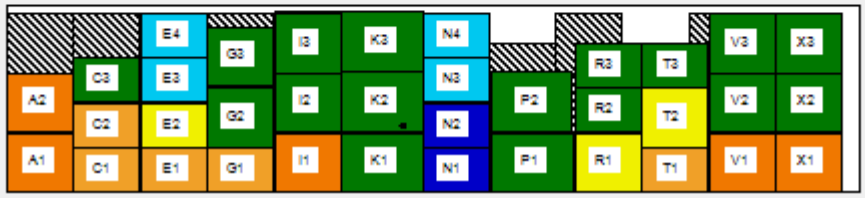
Bin type	Bin dimensions	xMax (mm)	xMax Ratio	weight (kg)	weight Ratio	volume (m3)	volume Ratio
UT12X	5899(L)x2350(l)x2393(H)	5890	99,85 %	12705,20	45,38 %	29,60	89,24 %

1 m	1 m	1 m	1 m	1 m	0,90 m
2,25 T	1,59 T	1,79 T	2,43 T	2,27 T	2,37 T



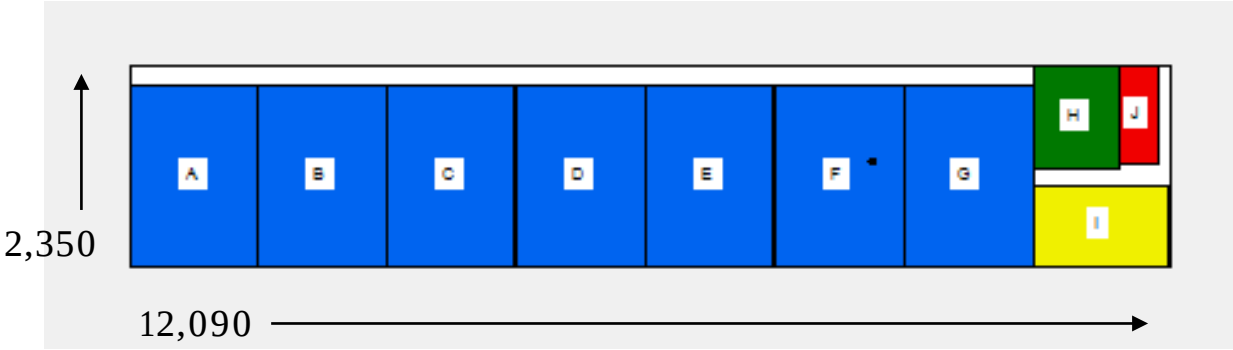
Bin type	Bin dimensions	xMax (mm)	xMax Ratio	weight (kg)	weight Ratio	volume (m3)	volume Ratio
CT40H	12090(L)x2350(l)x2680(H)	11860	98,10 %	24780,39	93,87 %	63,40	83,27 %

1 m	1 m	1 m	1 m	1 m	1 m	1 m	1 m	1 m	1 m	1 m	1 m	0,09 m
2,98 T	2,38 T	2,29 T	2,02 T	2,39 T	1,67 T	1,43 T	1,41 T	1,91 T	2,32 T	2,13 T	1,86 T	0,00 T

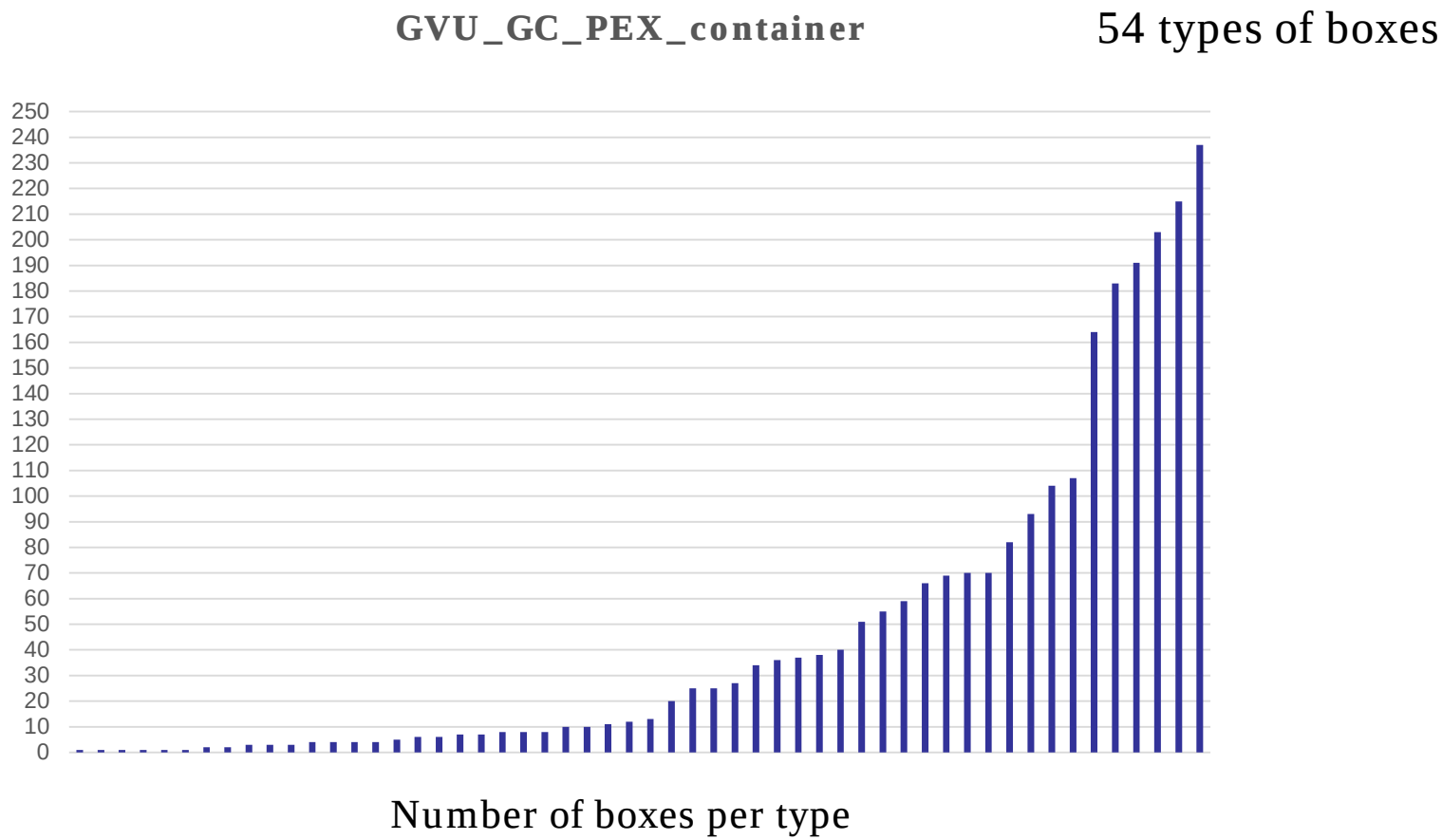


Very big and heavy boxes

- AVF_RIR_STK_container
  - 1307 kg. 2270x1170x1270
  - 38 boxes with more than 1000 kg.



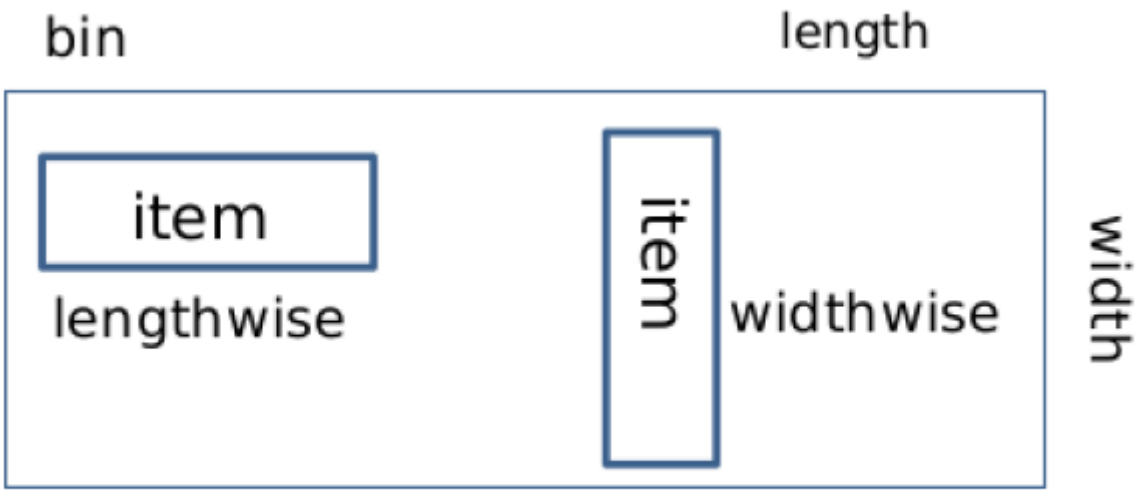
Extremely varying demand



Orientation constraints

- Items may be rotated only in one dimension (horizontal), so the bottom of a box always remains the bottom.

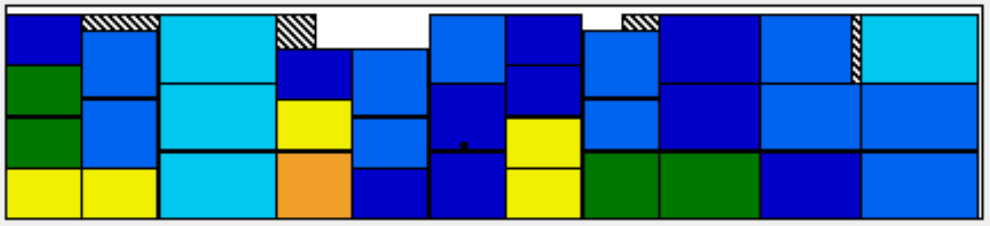
- Some items cannot be rotated, so they will be fixed lengthwise or widthwise (by convention, length is larger then width)

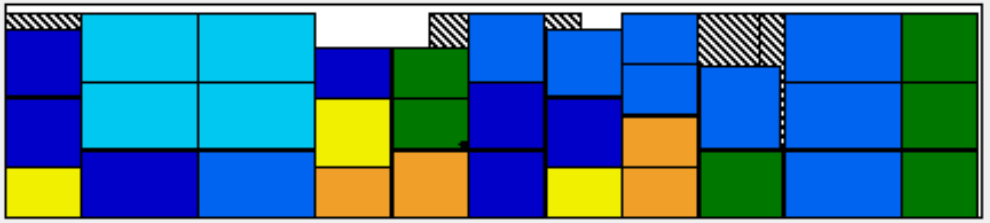


Very strict conditions concerning the weight

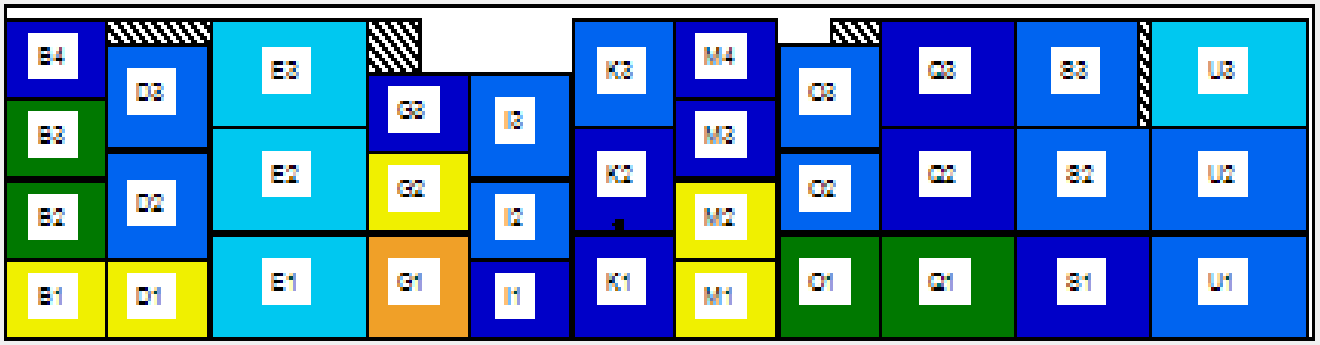
- Maximum density for non-metallic stacks
- Layers in a stack have to be sorted by decreasing weight
- Maximum weight supported by the items in the bottom stack

	< 50 Kg/m²
	50 to 100 Kg/m²
	100 to 200 Kg/m²
	200 to 300 Kg/m²
	300 to 400 Kg/m²
	400 to 600 Kg/m²
	600 to 1000 Kg/m²
	> 1000 Kg/m²





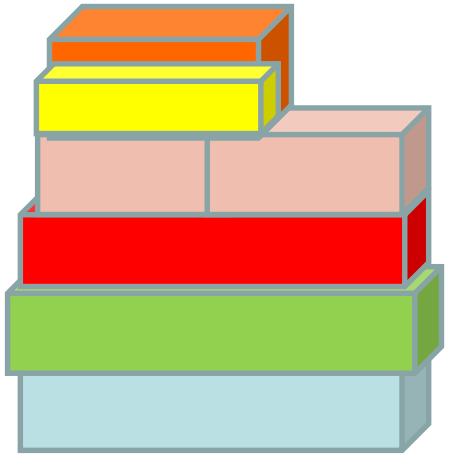
Strict conditions for building layers and stacks



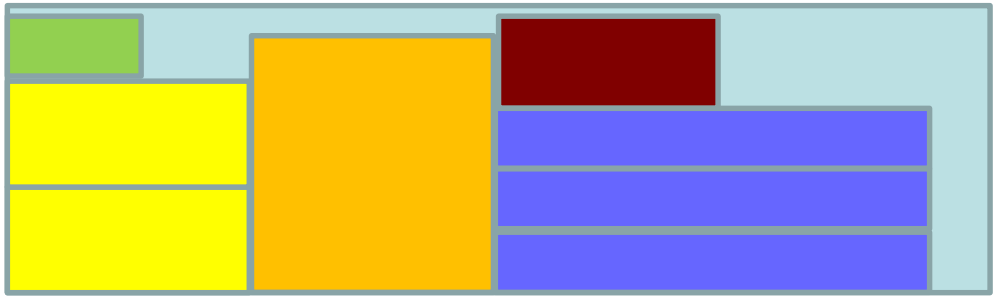
Two layers of the stack must not differ from more than a given ratio δ in the two horizontal dimensions ($\delta = 0.05$)

Special conditions for the top layer

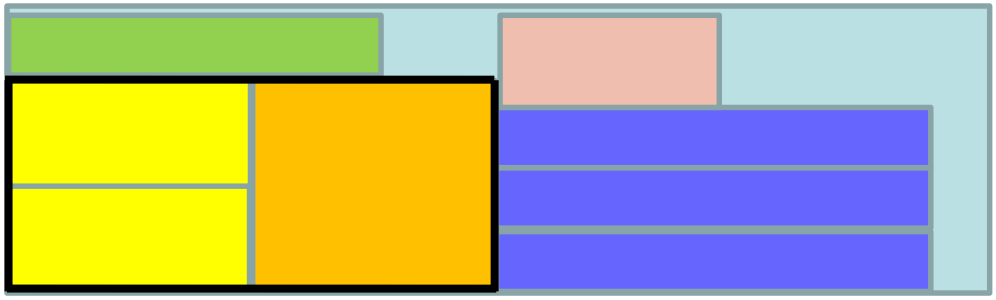
- Built when the stack is finished



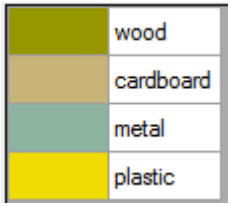
- Built when the bin is finished

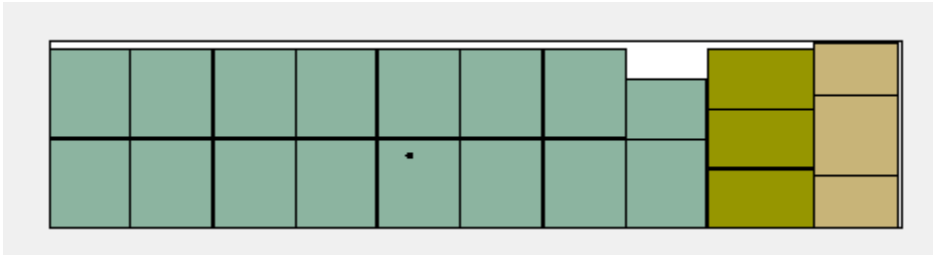


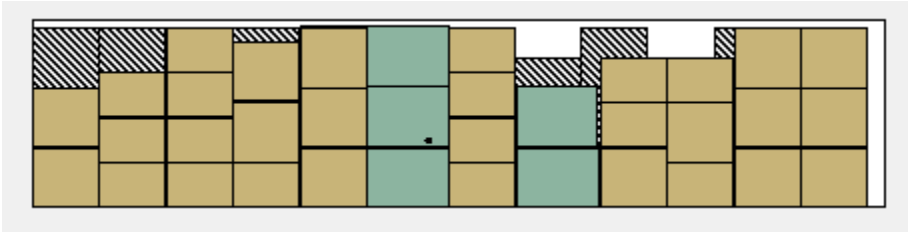
- Merging stacks

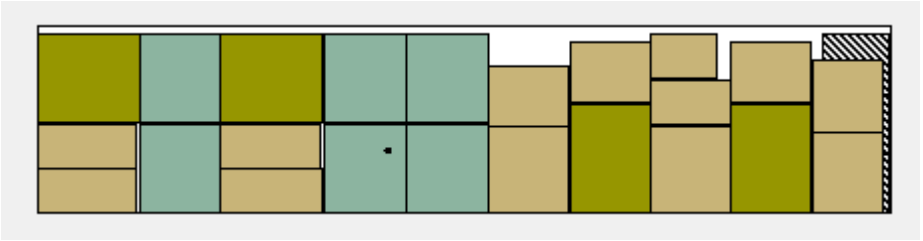


Strict separation of metal and cardboard/wood boxes









Don’t send half empty trucks!

- Bin 0 : Boxes that are not sent

  - For a given product with *k* items, there cannot be more than

    $$\lfloor k * \gamma \rfloor \quad \gamma \in [0,1] \quad (\gamma = 0.9)$$

  - Bin 0 defined as the one with the smallest volume used
    (known at the end of the loading process)